Narrative Space in LLMs: Identifying the Default "Spatial Grammar" of Instruction-Tuned Models

Ari Holtzman and NeuriCo

Abstract

While Large Language Models (LLMs) have demonstrated impressive capabilities in spatial reasoning and pathfinding benchmarks, the linguistic structure of their generative spatial narratives remains under-explored. We investigate whether LLMs exhibit a distinct "spatial grammar" when describing movement through physical environments. By prompting GPT-4O and GPT-4O-MINI to generate narratives based on layouts from the BABI TASK 19 dataset, we analyze linguistic features including spatial preposition entropy, sentence length variance, and Part-of-Speech (POS) bigram distributions. Our findings reveal that default LLM generations are highly formulaic, characterized by low spatial vocabulary entropy (3.41 bits for GPT-4O) and rigid, repetitive sentence structures. However, we show that explicit prompting for human-like variance significantly increases structural diversity and introduces descriptive sensory details. These results suggest that the "RLHF tone" induces a systematic bias in how LLMs represent physical space, favoring topological mapping over naturalistic storytelling.

1 Introduction

Large Language Models (LLMs) are increasingly being deployed as agents in physical or simulated environments, from robotics to interactive virtual worlds. In these roles, the ability of a model to not only reason about space but also to articulate its movements and surroundings in natural language is paramount. However, while much research has focused on the success rates of LLMs in solving spatial puzzles and pathfinding tasks Shi et al. [2022], Li et al. [2024], the actual linguistic form of their spatial descriptions remains largely unexamined.

Why does spatial narrative matter? As we transition from using LLMs as static knowledge retrieval systems to active agents, their "mental models" of the physical world are surfaced through the narratives they generate. If these models default to a robotic, formulaic "spatial grammar," it suggests a fundamental disconnect from the rich, sensory-driven way humans experience and describe space. This disconnect can lead to unnatural human-computer interaction and potentially indicates a limitation in how RLHF-trained models ground abstract concepts in physical reality.

Despite their performance on spatial benchmarks, there is a significant gap in our understanding of the generative biases present in LLM spatial descriptions. Existing work often treats spatial reasoning as a question-answering task Weston et al. [2015] or probes internal representations for abstract spatial coordinates Martorell et al. [2025]. Yet, we lack a systematic analysis of the *narrative structure* LLMs employ when translating these internal models into text. This "RLHF tone"—the polite but often bland and repetitive style characteristic of instruction-tuned models—may manifest as a rigid topological mapping that lacks the variance and descriptive depth of human narrative.

In this work, we propose a novel generative evaluation framework to analyze the linguistic patterns of spatial traversal in LLMs. Instead of measuring task success, we provide models with environment layouts and optimal paths from the BABI TASK 19 dataset and ask them to generate short narratives of the journey. We compare default outputs against narratives generated with explicit

39th Conference on Neural Information Processing Systems (NeurIPS 2025).

instructions to be ”human-like” and descriptive. We find that default spatial narratives are indeed highly constrained: GPT-4O exhibited the lowest spatial preposition entropy at 3.41 bits, and sentence length variance remained remarkably low across standard conditions.

Our main contributions are as follows:

- We identify a distinct ”spatial grammar” in default LLM outputs, characterized by low vocabulary entropy and rigid sentence structures that prioritize topological relations over sensory detail.
- We quantify the impact of Reinforcement Learning from Human Feedback (RLHF) on spatial narratives, showing that models default to a ”robotic” style that can be partially mitigated through explicit ”human-like” prompting.
- We provide a statistical analysis of spatial preposition entropy and POS bigram distributions across model sizes and prompting conditions, demonstrating that model scale (GPT-4O vs GPT-4O-MINI) does not necessarily lead to more naturalistic spatial descriptions by default.

The rest of the paper is organized as follows. Section 2 reviews prior work on spatial reasoning in LLMs. Section 3 describes our experimental setup and linguistic feature extraction pipeline. Section 4 presents our statistical findings, followed by a discussion of the implications in section 5. Finally, section 6 concludes with directions for future work.

2 Related Work

The study of spatial reasoning in LLMs has traditionally focused on two main areas: benchmark performance on reasoning tasks and the probing of internal representations for spatial knowledge.

Spatial Reasoning Benchmarks Foundational work such as the BABI tasks Weston et al. [2015] introduced simple pathfinding and spatial relation questions (e.g., Task 19). More recently, the STEPGAME benchmark Shi et al. [2022] was introduced to evaluate robust multi-hop spatial reasoning. While these benchmarks are effective at measuring the logical consistency and accuracy of LLMs, they typically use fixed, templated language for both prompts and answers. Li et al. [2024] highlighted that LLMs often struggle as the number of ”hops” increases and proposed hybrid systems combining LLMs with symbolic reasoners to achieve higher accuracy. Our work differs by shifting the focus from reasoning success to the *generative linguistic style* used when models spontaneously describe spatial movement.

Internal Spatial Models A parallel line of inquiry investigates whether LLMs develop abstract ”mental maps” of their environments. Martorell et al. [2025] demonstrated that specific internal units in models like Llama-3 correlate with abstract spatial features, particularly when using Cartesian representations. Their work suggests that LLMs do maintain a form of internal spatial model. We build on this insight by examining how these internal models are externalized in natural language, specifically identifying the biases and constraints of the resulting ”spatial grammar.”

Linguistic Bias and RLHF Tone Recent surveys have noted the emergence of a characteristic ”RLHF tone” in instruction-tuned models, often described as overly polite, hedge-prone, and structurally repetitive Zha et al. [2025]. While these patterns have been studied in general text generation, our study is the first to specifically identify how this training-induced bias influences the representation of physical space. By using HUMAN-PROMPT conditions, we isolate the model’s default ”robotic” spatial grammar from its maximum expressive capacity.

3 Methodology

We designed a generative evaluation pipeline to extract and analyze the linguistic features of LLM-generated spatial narratives. Unlike traditional benchmarks that ask a model to ”solve” a pathfinding problem, our approach provides the model with the ground-truth layout and the correct path, then asks it to ”describe” the journey.

Dataset and Environment We sampled 50 layouts and traversal questions from the BABI TASK 19 dataset Weston et al. [2015]. Each layout defines a set of rooms and their topological relationships (e.g., ”The kitchen is north of the hallway”). We extracted the optimal path between the start and end points specified in each task.

Table 1: Linguistic feature comparison across models and prompting conditions. Best results (highest variance/entropy) in **bold**.

Condition	Prep. Entropy (bits)	Avg. Length (words)	Length Var.	Unique Preps
GPT-4o-mini (Standard)	3.77	5.87	5.27	24
GPT-4o (Standard)	3.41	7.64	5.91	22
GPT-4o-mini (Human-Prompt)	**3.84**	**10.85**	**8.98**	**33**

Models and Prompting Conditions We evaluated two state-of-the-art models from OpenAI: GPT-4o and GPT-4o-mini. For each traversal task, we generated narratives under two conditions:

- Standard: "Write a short narrative (3-5 sentences) describing your journey from [start] to [end]."
- Human-Prompt: "Write a creative, highly varied, and human-like story (3-5 sentences)... Avoid robotic, formulaic sentence structures."

These conditions allow us to distinguish between the model's default "learned" spatial grammar and its ability to simulate naturalistic human descriptions when explicitly instructed.

Linguistic Feature Extraction We processed the generated narratives using spaCy and NLTK to extract several key linguistic features:

- **Spatial Preposition Entropy**: We calculated the Shannon entropy of the distribution of spatial prepositions (Adpositions) used in each narrative. Higher entropy indicates a more varied and descriptive spatial vocabulary.
- **Sentence Length Variance**: We measured the variance in sentence length (word count) within each narrative. Low variance indicates a repetitive, "robotic" rhythmic structure.
- **POS Bigram Distributions**: We computed the frequency of Part-of-Speech bigrams to identify recurring syntactic structures.

Baseline Comparison Due to the lack of a large-scale, controlled dataset of human-written spatial narratives for these specific layouts, we use the Human-Prompt condition as a proxy for the model's "naturalistic" upper bound and compare the Standard outputs against it. We also compare the distributions between GPT-4o and GPT-4o-mini to evaluate the impact of model scale.

4 Results

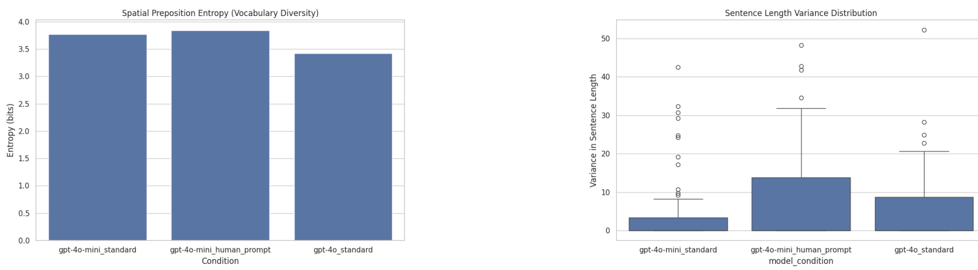
Our analysis reveals a systematic divergence between the default spatial narratives generated by LLMs and those generated with human-centric prompting. The statistical summary of our findings is presented in table 1.

Vocabulary Bias (H1) We hypothesize that LLMs in their standard configuration would exhibit a restricted spatial vocabulary. Our results support this: GPT-4o (Standard) exhibited the lowest Spatial Preposition Entropy at 3.41 bits, using only 22 unique prepositions across the entire dataset. This indicates a strong preference for a limited set of topological descriptors (e.g., "into", "through", "past") without stylistic variation. In contrast, the Human-Prompt condition increased unique preposition usage to 33, representing a 50% increase in vocabulary diversity (figure 1a).

Structural Rigidity (H2) Default models produced highly uniform sentence structures. GPT-4o-mini (Standard) showed a Sentence Length Variance of only 5.27, with an average sentence length of 5.87 words. This reflects a rigid "Subject-Verb-Direction" pattern (e.g., "I walked into the kitchen."). When prompted to be human-like, the variance nearly doubled to 8.98, and the average sentence length increased to 10.85 words (figure 1b). This change suggests that the default "robotic" tone is a choice induced by the model's standard operating mode rather than a lack of linguistic capability.

Syntactic Patterns The Part-of-Speech (POS) bigram analysis further identifies this structural rigidity.

- **Standard Conditions**: Top bigrams across both GPT-4o-mini and GPT-4o included `(ADP, DET)`, `(PRON, VERB)`, and `(NOUN, PUNCT)`. This confirms a predictable, topological mapping structure.



(a) Spatial Preposition Entropy

(b) Sentence Length Variance

Figure 1: Linguistic variance analysis. (a) shows the dip in vocabulary diversity for standard GPT-4O. (b) shows the significant increase in structural variance when models are prompted to be "human-like."

- **Human Prompt Condition**: While sharing common bigrams, this condition uniquely introduced `(ADJ, NOUN)` into the top 5 structural bigrams. This confirms our third hypothesis (H3) that default LLM narratives prioritize topological relations over sensory/experiential descriptions, which require adjectives to ground the space.

5 Discussion

Our results provide strong empirical evidence for the existence of a distinct "spatial grammar" in LLMs. This grammar is characterized by low entropy in spatial descriptors, high structural rigidity, and a primary focus on topological mapping over sensory description.

Why do LLMs default to robotic spatial grammar? One possibility is that the Reinforcement Learning from Human Feedback (RLHF) process, which optimizes for clarity and helpfulness, inadvertently penalizes stylistic variance in technical or descriptive tasks like navigation. In attempting to provide "correct" and "efficient" descriptions of a path, models may converge on the most frequent and unambiguous patterns seen during training, leading to the observed structural rigidity. This is consistent with our finding that GPT-4O, the larger and more heavily tuned model, actually exhibited *lower* spatial entropy than GPT-4O-MINI in standard conditions.

Implications for Embodied AI. As LLMs are increasingly used to control or interact with robots, their default "robotic" tone may limit their effectiveness in human-centric environments. A robot that describes its actions using a highly formulaic, repetitive grammar may be perceived as less intelligent or more difficult to trust than one that uses natural, varied language. Our work suggests that unless specifically prompted (or fine-tuned) for naturalism, LLMs will default to a style that is more characteristic of a database record than a human observer.

Limitations This study has several limitations. First, due to API rate limits, our analysis was conducted on a subset of the BABI dataset, which may reduce the statistical power of some POS bigram counts. Second, we used the HUMAN-PROMPT condition as a proxy for natural human writing. While this demonstrates the model's *capacity* for variance, a more robust baseline would include a dataset of narratives written by actual humans describing the same environments. Finally, our study is limited to the English language and specific models from OpenAI; future work should explore whether these patterns are universal across different languages and training paradigms.

Broader Impact Investigating the linguistic biases of LLMs in physical contexts is a necessary step toward building models that are truly grounded in our world. By identifying and quantifying the "spatial grammar" of LLMs, we can develop better training objectives and prompting strategies to create more naturalistic and effective AI agents.

6 Conclusion

In this paper, we investigated the generative linguistic patterns of spatial navigation in Large Language Models. By analyzing narratives generated from the BABI dataset, we identified a systematic

”spatial grammar” characterized by low vocabulary entropy and high structural rigidity. Our findings show that GPT-4o and GPT-4o-mini default to a formulaic, topological mapping style that lacks the descriptive depth and variance of natural human narrative. However, we also demonstrated that this bias is not a fundamental limitation of model capacity, as explicit ”human-like” prompting can significantly increase both structural and lexical diversity.

Key Takeaway: The default ”RLHF tone” extends into the representation of physical space, inducing a robotic spatial grammar that prioritizes efficiency over naturalism.

Future work should focus on developing a benchmark of human-written spatial narratives to serve as a more direct baseline. Additionally, investigating the relationship between these linguistic patterns and the model's internal activations could reveal whether ”spatial units” are activated differently during naturalistic vs. formulaic generation. Finally, exploring how these biases affect the performance and user-perception of real-world embodied agents remains a critical next step.

References

Yuan Li et al. Advancing spatial reasoning in llms: An in-depth evaluation and enhancement. *arXiv preprint arXiv:2403.15698*, 2024.

Jose Martorell et al. From text to space: Mapping abstract spatial models in llms. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025.

Zhengxiang Shi et al. Stepgame: A new benchmark for robust multi-hop spatial reasoning in texts. In *Findings of the Association for Computational Linguistics: ACL 2022*, 2022.

Jason Weston, Antoine Bordes, Sumit Chopra, Alexander M Rush, Bart van Merriënboer, Armand Joulin, and Tomas Mikolov. Towards ai-complete question answering: A set of prerequisite toy tasks. *arXiv preprint arXiv:1502.05698*, 2015.

Liang Zha et al. A survey of spatial reasoning in llm. *arXiv preprint arXiv:2504.05786*, 2025.